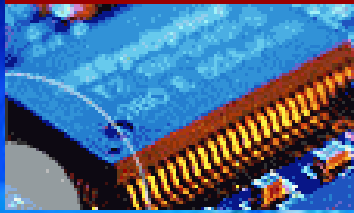
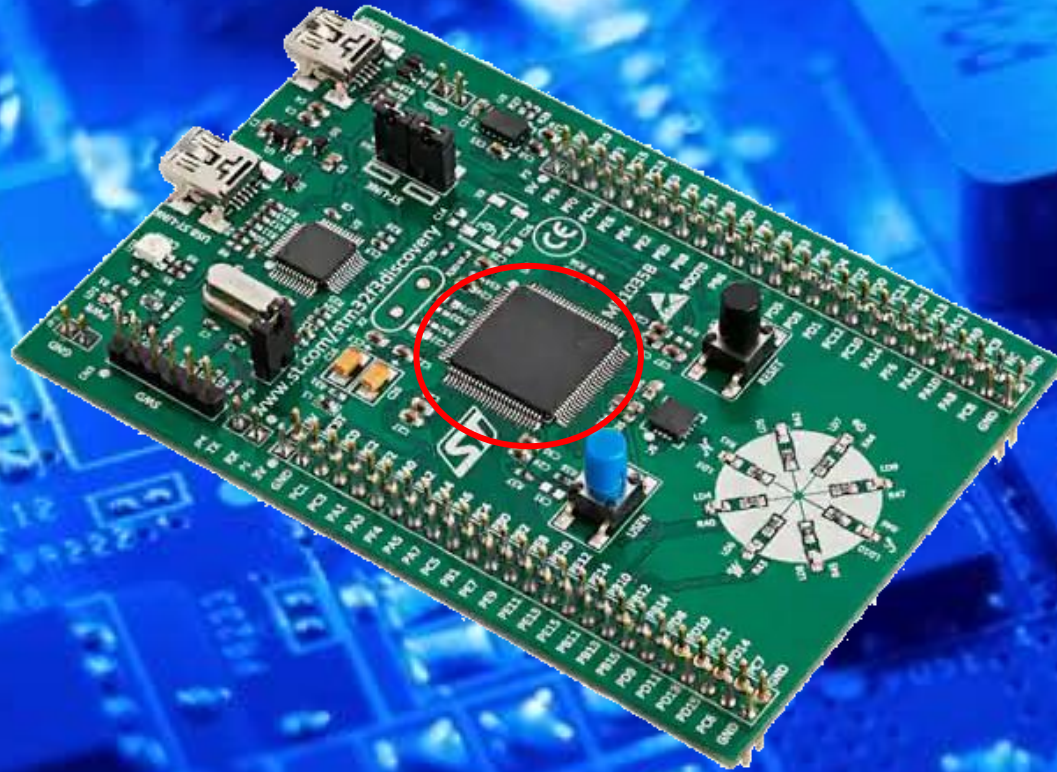


EEE3096S



Embedded Systems II



ADC Metrics (ADCs part 2 of 2)

L17

Embedded Systems II

Dr Simon Winberg



Added some clickable audio machine-gen annotations, same as what comments indicate.



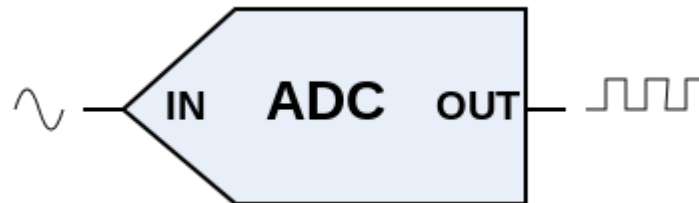
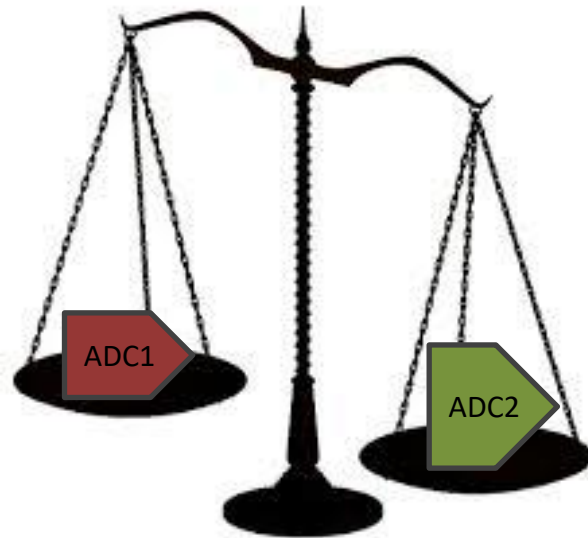
Electrical Engineering
University of Cape Town

Outline of Lecture

- ADC Resolution
- ADC/DAC Quantization
- Static Metric of an ADC
- Dynamic Metrics of an ADC



ADC Metrics



Analogue to Digital Convertors (ADCs)



ADC Resolution



Resolution (in bits) : number of bits produced

Resolution Q (in volts): difference between two input voltages causing the output to be incremented by 1 *

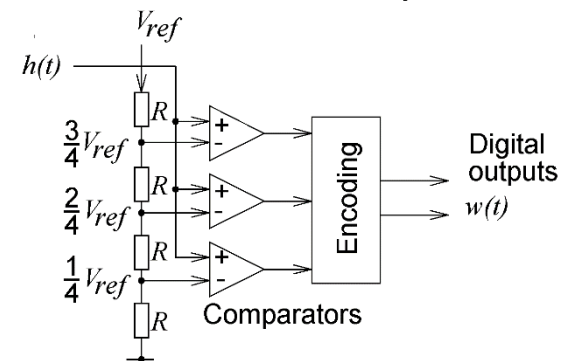
$$Q = \frac{V_{FSR}}{n} \quad (\text{quantitation factor calculation})$$

where:

Q : resolution in volts per step
 V_{FSR} : difference between largest and smallest voltage
 n : number of voltage intervals

Example:

$Q = V_{ref}/4$ for this flash ADC example



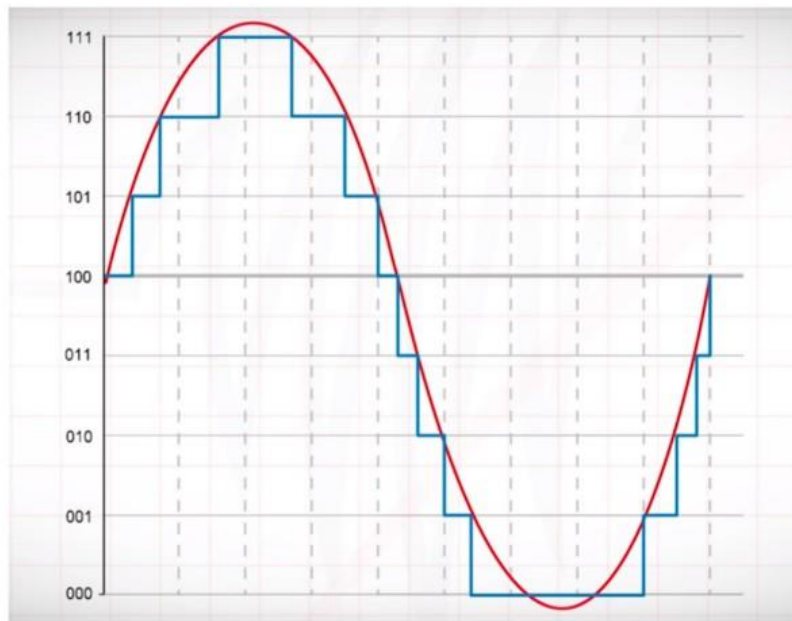
* i.e. this can be found practically using a voltage source that can be finely adjusted and see when the ADC value changes by 1. Obviously would need a little program running to somehow show the ADC's current reading.

Information source [1]: https://en.wikipedia.org/wiki/Analog-to-digital_converter

ADC/DAC Quantization

Definition of Quantization:

Mapping analogue voltages to digital codes



3-bit ADC Digital Output Code Table

Analog Input Range (Normalized to V_{REF})	Digital Output Code
7/8 $\rightarrow$ 8/8	111
6/8 $\rightarrow$ 7/8	110
5/8 $\rightarrow$ 6/8	101
4/8 $\rightarrow$ 5/8	100
3/8 $\rightarrow$ 4/8	011
2/8 $\rightarrow$ 3/8	010
1/8 $\rightarrow$ 2/8	001
0 $\rightarrow$ 1/8	000

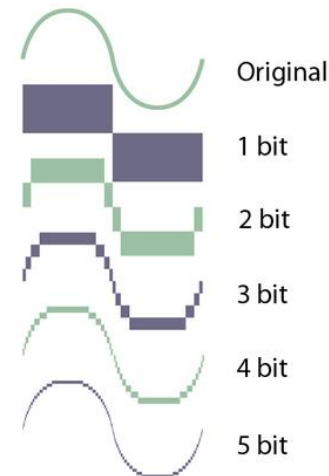
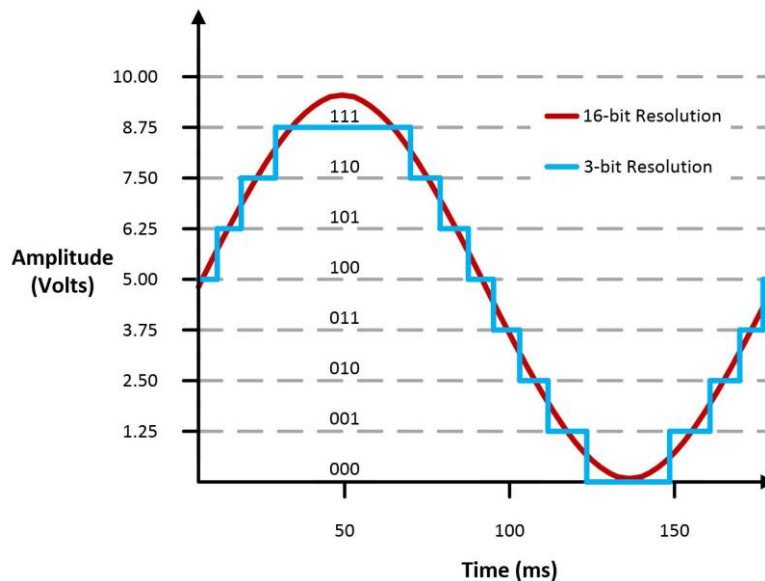
}1LSB Difference

Recommended video: <https://www.youtube.com/watch?v=xX6ifWthjDA>



Resolution of an ADC

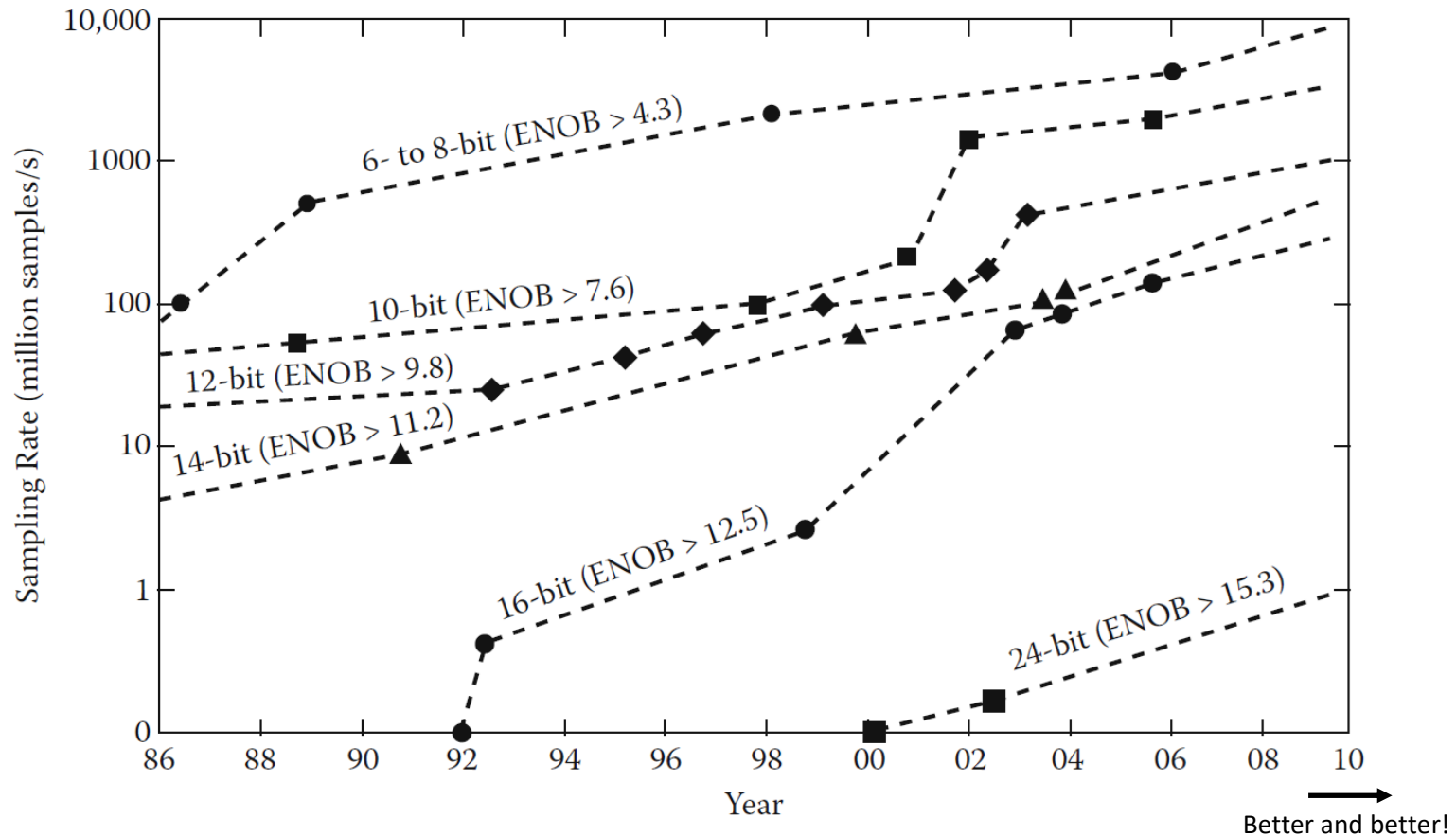
Number of Unique Codes = 2^N , where N is the number of bits



This answers the question “how many voltage levels can your ADC or DAC quantize?” i.e. the answer to this question would be 2^N in this case.

The resolution of an ADC or DAC is the number of bits, i.e. N , *not* 2^N .

Trends over the years in resolution



Static Metrics

The Static Metric of an ADC

- The most simple but also very important ADC performance metrics is the
 - Static, DC (direct current, or zero-frequency constant input) performance of the devices.
 - i.e. does it read 0 when the input is 0V? *

* Note: you may need to interpret this differently depending on how your ADC is configured. For example, sometimes the configuration is such that 0 is output for $-V_{max}$ and $2^N - 1$ for $+V_{max}$... in which case we'd need to see what gives for 0V (and you might have to look at the average of the middle codes to be more fair).

Static Metric: Offset Error

OFFSET ERROR:

The offset error of an ADC, which is similar to the offset error of an amplifier, is defined as a deviation of the ADC output code transition points that is present across all output codes.

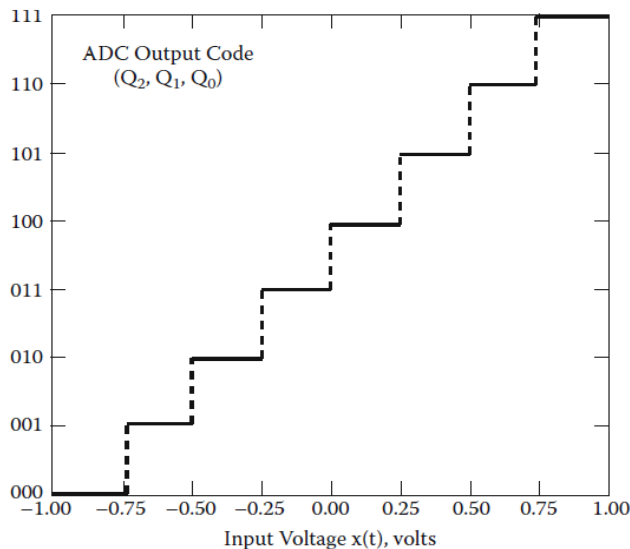


FIGURE 7-2 Conceptual ADC transfer function.

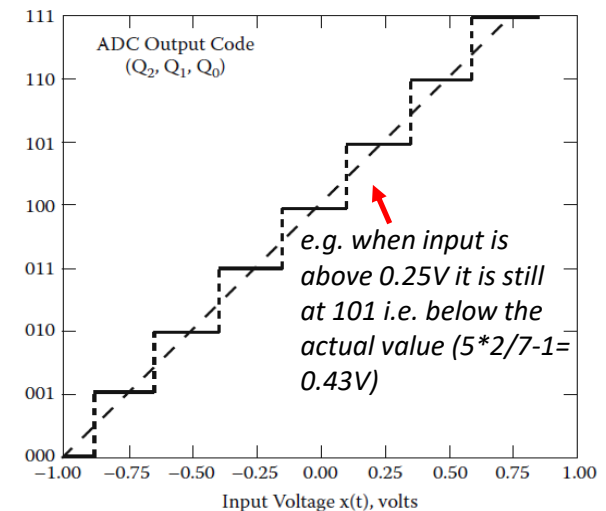
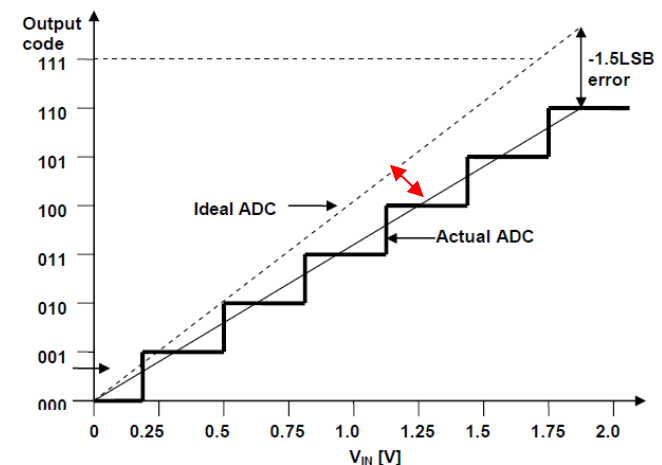


FIGURE 7-3 Ideal ADC transfer function.

Static Metric: Gain Error

GAIN ERROR:

- The gain error of an ADC is similar to the gain error of an amplifier.
- Assuming ADC's offset error removed, then:
 - The gain error determines the amount of “rotational” deviation away from the ADC's ideal transfer function slope (i.e., the dashed diagonal line of the figure).
- Once the gain error has been characterized, it may be possible to compensate for this error source.



Dynamic Metrics

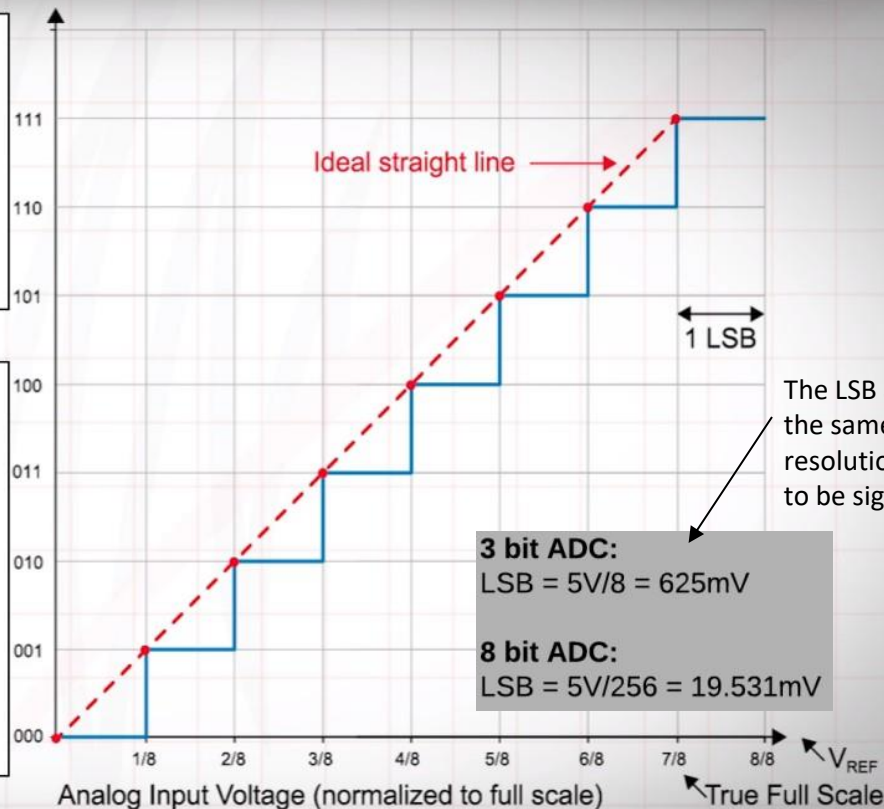
ADC & DAC Metric: LSB accuracy

This metric is used to compare how a LSB change in the code relates to the step change between the measured voltage V_{ref} (blue line) and the desired ideal transfer function (dotted line). The LSB accuracy may not be consistent for all codes; it may change for different V_{ref} s (e.g. get wider, a bigger jump, as the voltage increases).

$$\begin{aligned} \text{LSB} &= \frac{V_{ref}}{2^n} \\ &= \frac{8}{2^3} = 1 \text{ V} \end{aligned}$$

$$\begin{aligned} \text{LSB} &= \frac{V_{FSR}}{2^n - 1} \\ &= \frac{8}{2^3 - 1} = 1.14 \text{ V} \end{aligned}$$

$$V_{FSR} = V_{ref} - 1 \text{ LSB}$$



V_{FSR} : difference between largest and smallest voltage

n : number of voltage intervals

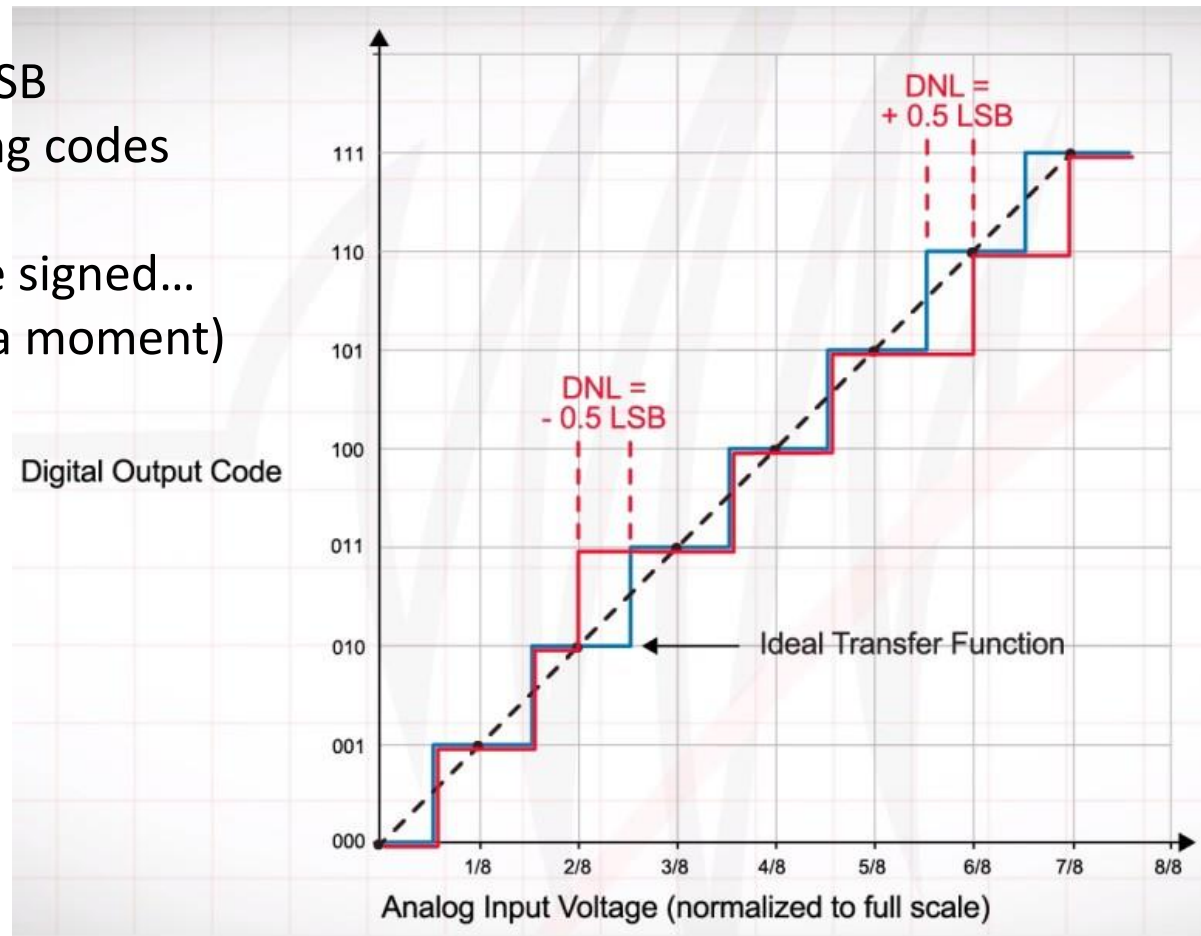
ADC & DAC Metric: Linearity (DNL)

Differential Non Linearity (DNL):

The deviation between actual steps and ideal steps

DNL must be $< \frac{1}{2}$ LSB
to ensure no missing codes

(The DNL should be signed...
we will see why in a moment)



ADC & DAC Metrics: Linearity (INL)

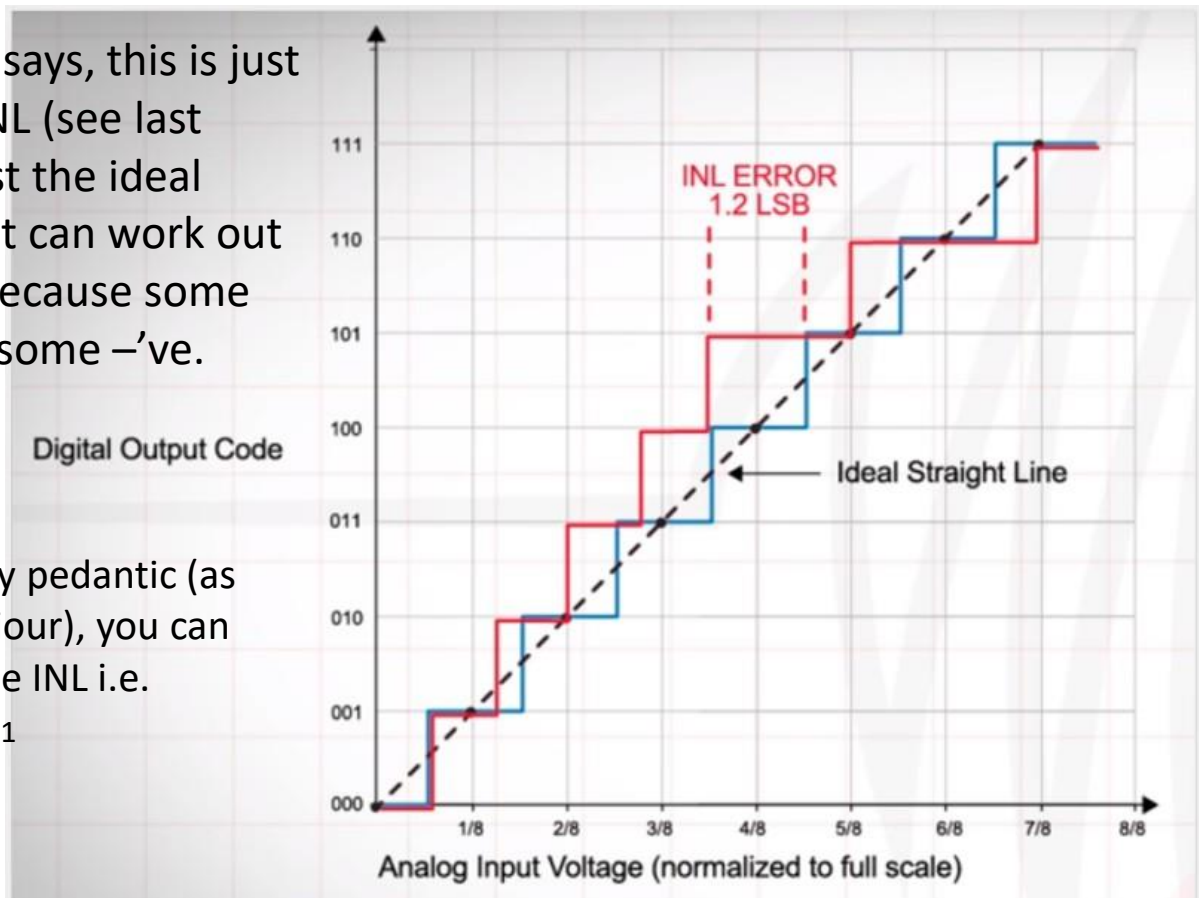
Integral Non Linearity (INL):

Integration of DNL error (accumulation of all DNL)

As the description says, this is just a sum of all the DNL (see last slide) errors against the ideal transfer function. It can work out to a low number because some DNLs may be +ve some -ve.

If you want to be very pedantic (as in type-A anal behaviour), you can work out the absolute INL i.e.

$$\sum |DNL_i| \text{ for } i = 0 \text{ to } (2^N)-1$$



Information source and recommended video:

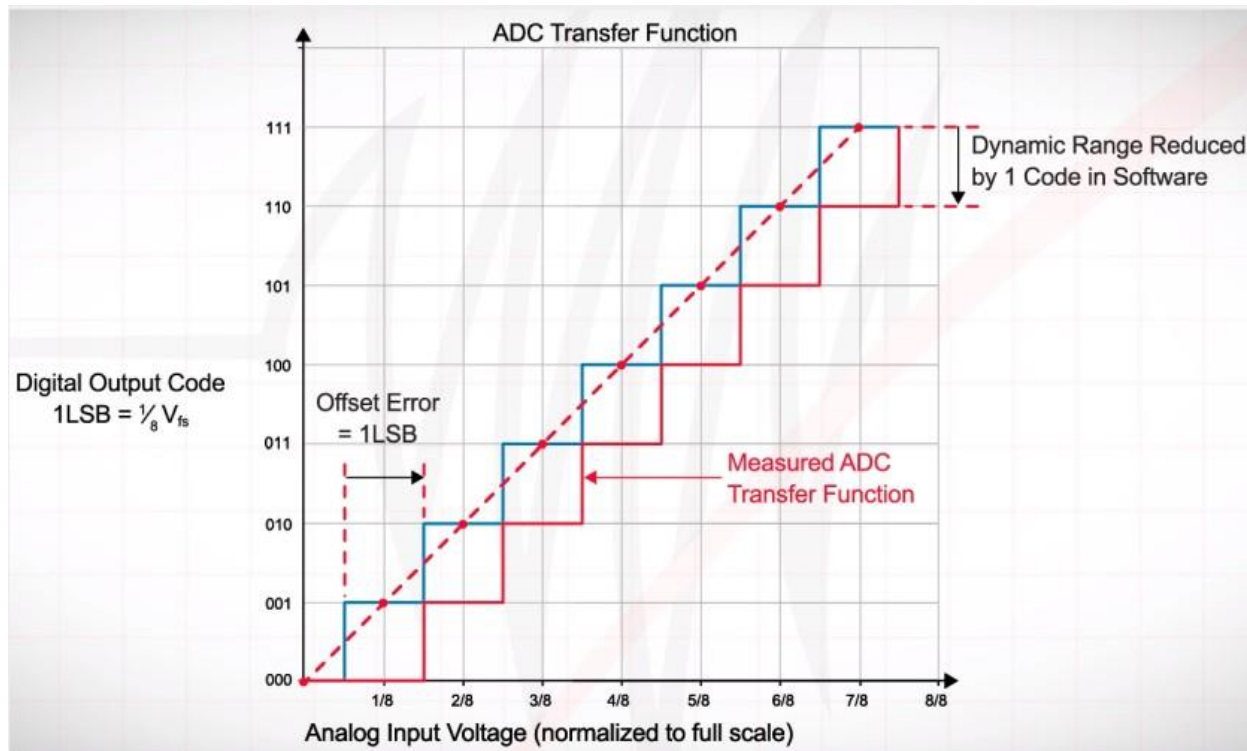
<https://www.digikey.com/en/videos/m/microchip/adc-aspects---episode-1---quantization>

Offset error (recap)

Remember from earlier slide:

Offset error = Deviation of the output

Can easily be corrected using the microcontroller (reduces range of an ADC)



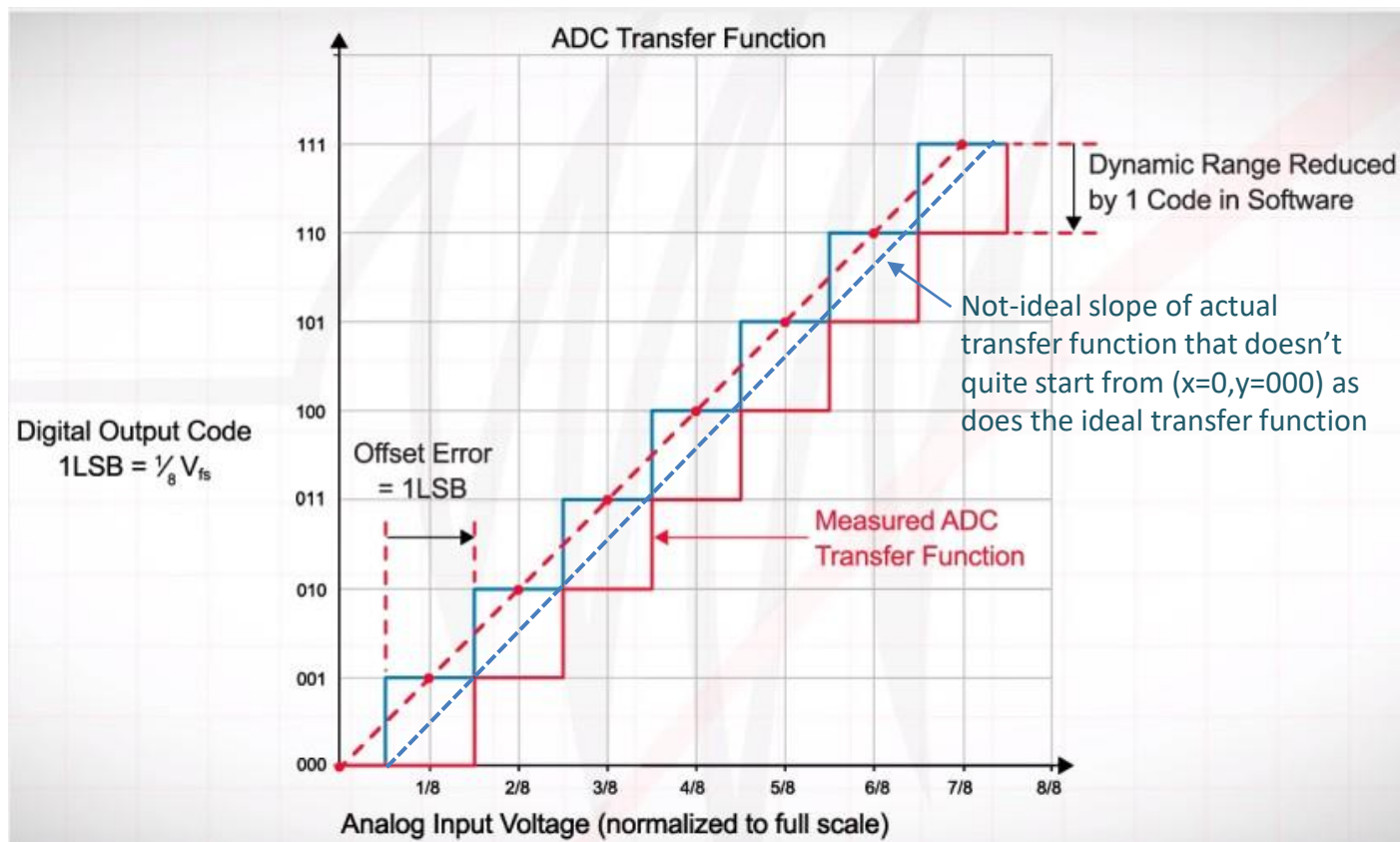
Offset error not to be confused with full scale error...

Full scale Error

Definition Full scale error:

Offset error at the last transition

- ADC reaches its highest code before reaching its highest voltage
- Caused by both offset error + non-ideal slope of the transfer function



Dynamic Measures:

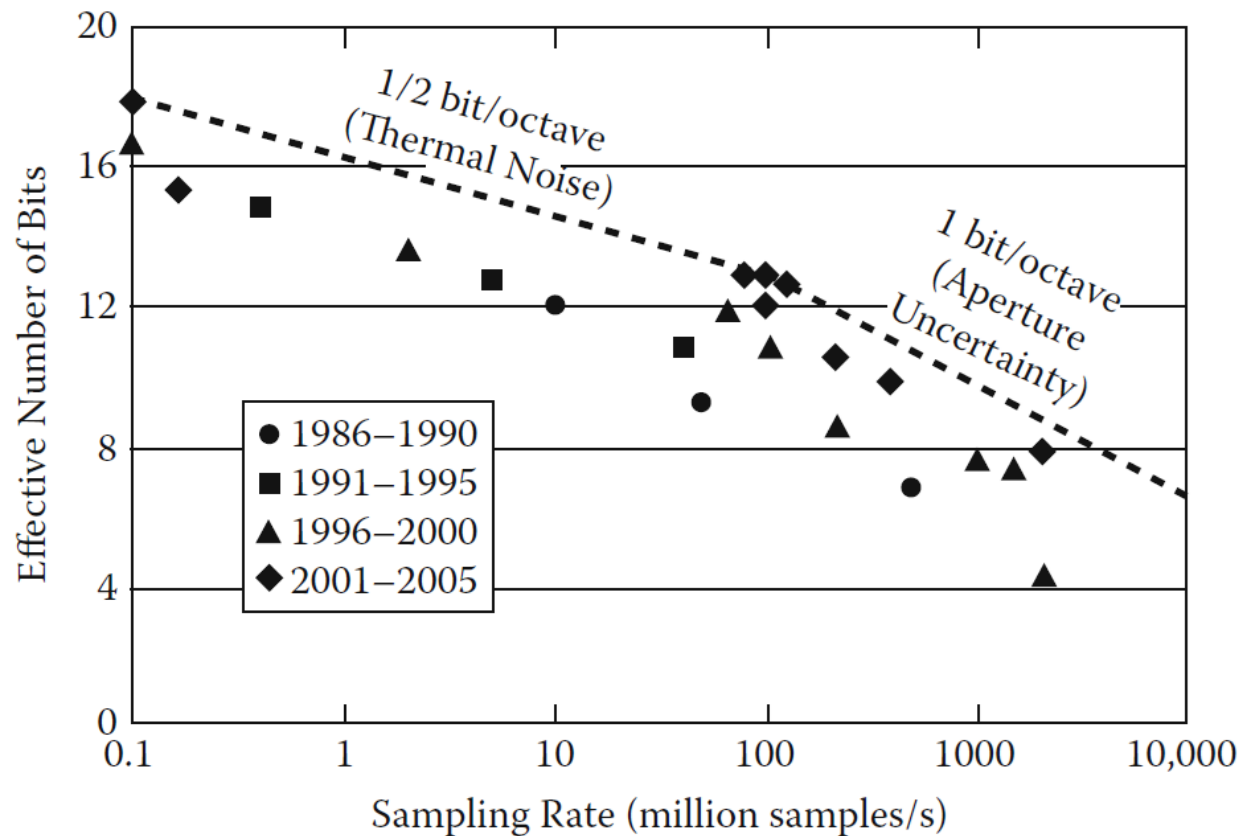
Effective Number of Bits (ENOB)

- Effective number of bits (ENOB)
 - The resolution of an ADC is specified by the number of bits used to represent the analogue value, in principle giving 2^N signal levels for an N-bit signal.
 - The ENOB indicates how many bits effectively contribute towards representing the sampled input.

Bad ENOBs are like bad apples for an embedded engineer (they can spoil the rest of your processing package)

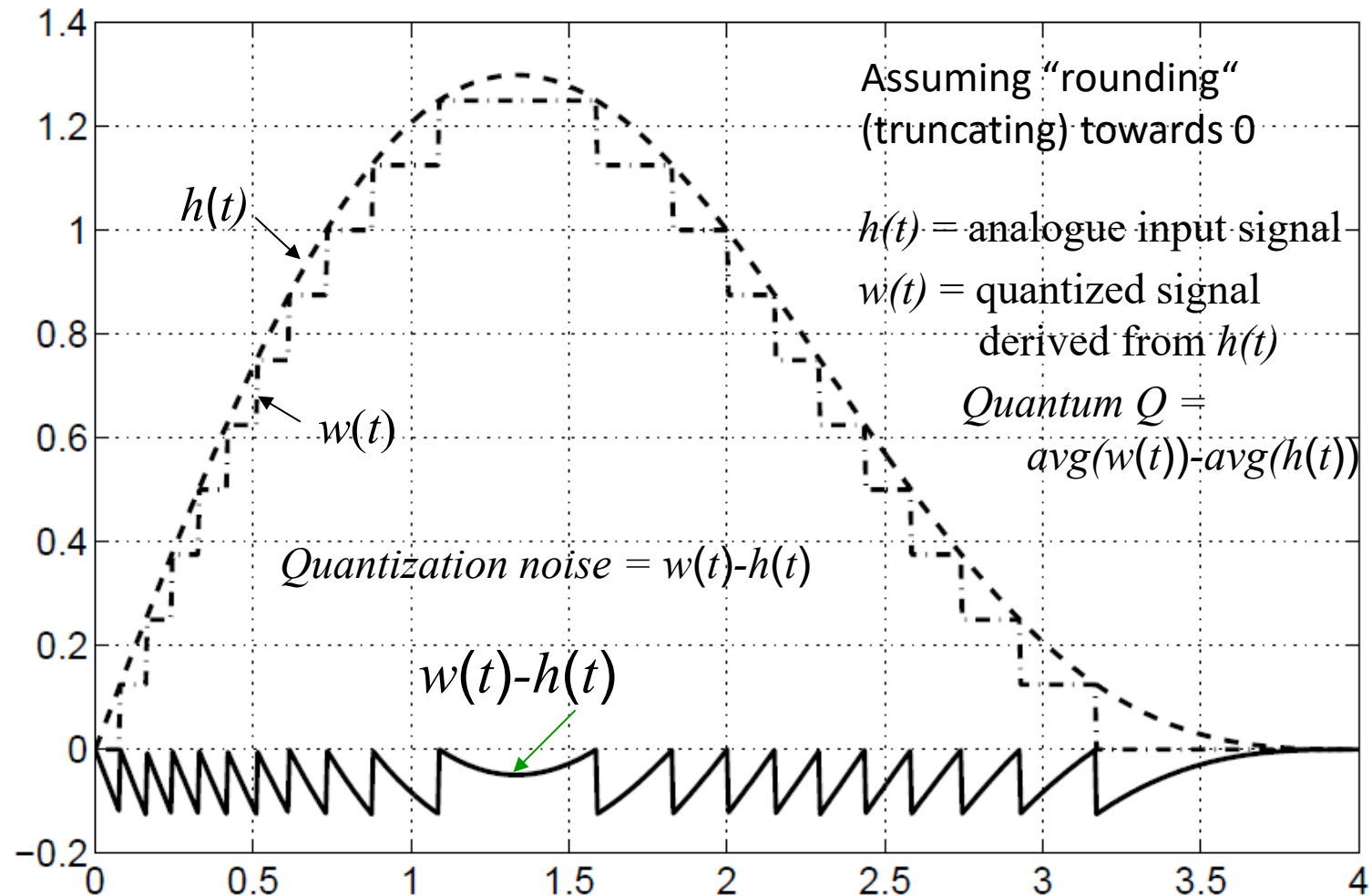


Trends in ENOB vs Speed



Quantization Noise

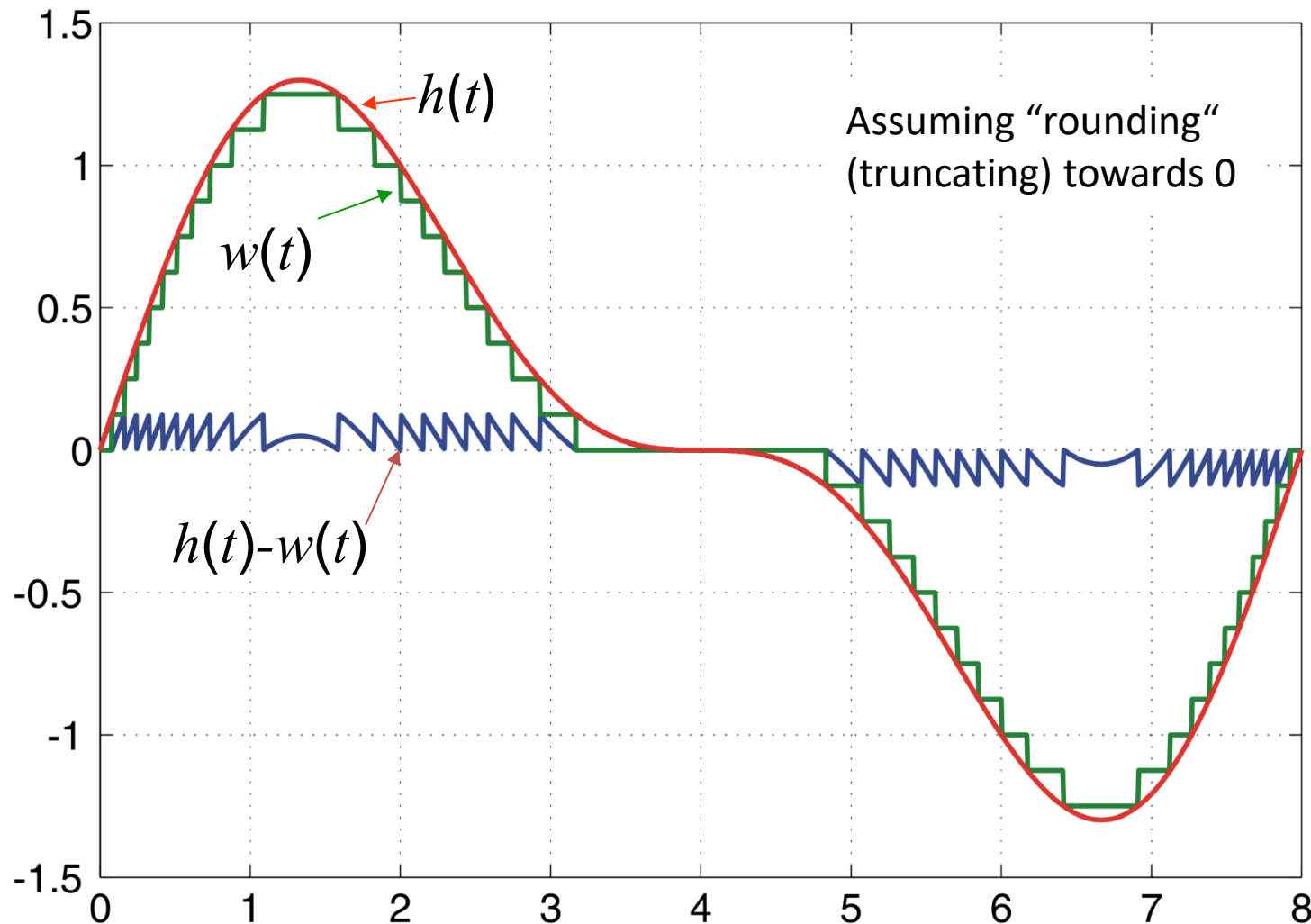
Not examined
(in 2025)



So, if I asked you to graph how the difference between the input signal (i.e. $h(t)$) and the quantization transfer (i.e. $w(t)$) looks you would be able to show that pretty easy now.

Quantization Noise

Not examined
(in 2025)



$h(t)$ = analogue input signal $w(t)$ = quantized signal derived from $h(t)$

Quantization Noise Impacts SNR

- Remember what we said about Shannon and channel capacity. Part of that exercise was working out the SNR. The quantization noise adds to other noise that may be hindering the channel.

- SNR = signal/noise

$$\text{SNR in dB} = 10\log_{10}(\text{signal/noise})$$

$$\text{SNR in dB} = 20\log_{10}(V_{\text{signal}}/V_{\text{noise}}) \quad *$$

$$\text{SNR}_Q \text{ in dB} = 20\log_{10}(n/\alpha Q)$$

α characterizes how well $w(t)$ tracks $h(t)$

$$\alpha=0 \rightarrow h(t) = w(t)$$

$$\alpha=1 \rightarrow w(t) \text{ a little off from } h(t)$$

clearly $\alpha=0 \rightarrow$ perfect (infinite) SNR $\alpha=1 \rightarrow$ strong signal

Not examined
(in 2025)

* For voltage the power is proportional to V^2 hence the x2 for in the 2nd equation (see net slide)

Reminder: dB for Amplitude vs Power

- The usual definition for dB, when using power (a current flowing in 1s at a voltage) is

$$\text{Power dB} = 10 * \log\left(\frac{\text{Power 2}}{\text{Power 1}}\right)$$

i.e. where you are comparing the **power** or **energy** of one signal to another.
This is because electric power is V*I Watts per second.

- But if you are looking at only one of the qualities, the amplitude or voltage that is being sent/received then clearly a compensation is made for that, for which the log is multiplied by 20 instead of 10 :

$$\text{Amplitude dB} = 20 \cdot \log\left(\frac{a_2}{a_1}\right)$$

Not examined
(in 2025)

Spurious-Free Dynamic Range (SFDR)

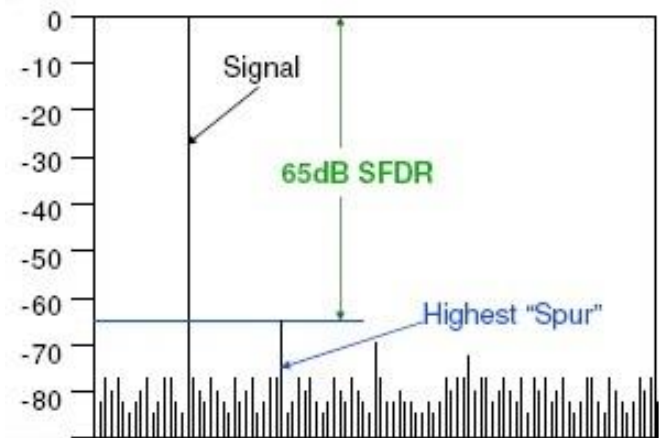
Not examined
(in 2025)

Spurious-free Dynamic Range (SFDR):

- This is a frequency-domain measurement that determines the **minimum signal level that can be distinguished from spurious components**, and includes all spurious components over the full Nyquist band regardless of their origin
- SFDR for a ADC typically decreases as sampling rate increases. (It is generally indicate as a power or S/N ratio value)

This can be calculated for sampling real systems by sampling a known sinusoidal reference (or carrier signal) with minimum interference, sample it for a period and perform a Fourier transfer to find the difference between the max. of the spurious signals and the full scale range (P_{FS} corresponds to the maximal power measured). Note: must put the spectrum in dBs i.e. $10\log_{10}(P_S/P_{REF})$.

Example of measuring SFDR



noise spurs are at the bottom of the plot

Done with ADC and DAC Theory!

Now we are DONE with ADCs 😊
& DACs
and



Full of ADC&DAC knowledge
for likely embedded systems needs



...and now you
can get comfy
in the cosy
digital world 😊



References

1. Analog-to-digital Converter
https://en.wikipedia.org/wiki/Analog-to-digital_converter
2. “High performance embedded computing handbook: a systems perspective”. Martinez, D., Bond, R. and Vai, M. eds. CRC Press (2008)

The Next Episode...

Lecture L18

L18: Computer Architecture

a. CPU Architecture (recap)

b. ALU and Datapath ← focused on

